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PROBLEM STATEMENTS

MANAGEMENT

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PS1

Case: The Dragon Passion Problem at Pulp Riot Juice Co.

Background

Pulp Riot Juice Co. is a fast-growing Indian maker of bottled fruit juices, known for unusual flavors and highprofile marketing. It operates two plants and one co-packer, and ships to nine depots that serve about 600 general-trade distributors, modern-trade retailers and quick-commerce dark stores across the country. Annual revenue is ₹1,200 crore, at a contribution margin of 32%. Its range has five flavors: 1. Mango Peach Basil (38% of summer volume) uses mango pulp from contracted processors, peach pulp from hill-state orchards and fresh basil from contract farms near the plant, pressed into extract on site. 2. Guava Chilli (20%) uses guava pulp from two regional suppliers. 3. Passionfruit Punch (16%) depends on a single hill-region cooperative. 4. Dragonfruit Bloom (14%) uses dragonfruit pulp that is 60% domestic and 40% imported, with a lead time of six to eight weeks. 5. Dragon Passion (12%), a new blend, uses both dragonfruit and passionfruit pulp. Peak summers (March to June) bring in 48% of annual revenue. The plants cannot add mid-season lines. The copacker can add up to 10% volume at a premium.

Key operating data:

• **Sales mix:** General trade accounted for 55% of summer volume, modern trade 25% and quick commerce 20%. The average realized price was ₹60 per bottle. • Planning: Forecasts are updated monthly, and forecast error averaged 30% at the flavor and depot level. Quick-commerce platforms send daily sell-through reports to marketing; planners receive consolidated sales data at month-end.

• **Production:** Summer volume was about 9.6 crore bottles. The two plants can make about 2.5 crore bottles a month and ran at 95% utilization through the summer. The co-packer's extra volume (up to 10%) costs 25% more per bottle than in-house production and needs two weeks' notice.

• **Sourcing:** Procurement targets are set on price per kilogram, and no second passionfruit supplier has been qualified. Imported dragonfruit pulp is ordered once a quarter. Marketing set Dragon Passion's launch date and informed procurement two weeks before launch.

• **Shelf life:** Frozen fruit pulp keeps for up to six months in plant cold storage. Bottled juice has a 90-day shelf life, and modern-trade and quick-commerce buyers reject stock with less than 60 days left. • Routes: Passionfruit pulp normally reaches the plants by road in three days. There is no alternative road route from the cooperative, and the highway has closed for at least a week in three of the last five summers. The plants held about two weeks of passionfruit pulp when the May closure began, and air freight was booked in its second week.

• **Depots:** The nine depots sit in four regions. Each depot holds about three weeks of cover for every flavor, based on the previous year's flavor mix, and allocations are reset monthly. Each depot reports its stock weekly in its own spreadsheet. A transfer between depots takes two to four days and costs about ₹1.50 per bottle.

In early March, Pulp Riot launched Dragon Passion. Two weeks after launch, an influencer's 30-second reel featuring the drink went viral, and quick-commerce orders rose fourfold over the next two weeks. Planners saw the spike only when distributors and retailers began reporting shortages. Then, in May, landslides closed the only highway from the passionfruit cooperative for three weeks. The company spent ₹3.2 crore air-freighting pulp at five times the normal cost, cut Passionfruit Punch production to protect Dragon Passion, and still ran out of both products. Meanwhile, some depots were overstocked with Mango Peach Basil and Guava Chilli, which later cost ₹6 crore in write-offs and discounts, while other regions ran short of the same flavors. About 8% of summer revenue was lost to stock-outs.

Within six weeks of the stock-outs, a larger rival launched its own dragonfruit-passionfruit drink and took the shelf space Pulp Riot couldn't fill.

The CEO’s verdict: “We didn’t lose to the competitor; we lost to ourselves.”

Problem Statement

Pulp Riot’s planning involves marketing, sales, procurement, supply chain and finance, and is largely manual.
Key challenges include:
• Demand changes were not reflected in plans until shortages appeared.
• Passionfruit pulp ran short during the peak, and summer plant capacity is fixed, with the co-packer adding only up to 10% at a premium.
• Some flavors and regions ran out while stock built up elsewhere, costing ₹6 crore in write-offs and discounts.
• Stock did not reach the depots and channels where demand was rising, and there is no general-trade outlet data.
• The May highway closure cut off passionfruit pulp for three weeks, and imported dragonfruit pulp takes six to eight weeks. Air freight cost ₹3.2 crore at five times the normal cost. Peak-season trade-offs are escalated to the COO, and the S&OP meeting is held monthly. Fifteen planners work in spreadsheets, there is no data or AI team, IT is run by vendors, and two earlier digital pilots stalled after go-live.

Task

Build a model or agent that gives Pulp Riot end-to-end visibility across demand, supply, inventory, distribution and transportation, identifies risks early and recommends how to mitigate them, so that last summer’s failures do not recur next summer, eight months away. The model can be built in any tool, but the full rollout of your solution must fit within a total first-year budget of ₹2.5 crore, covering all one-time build costs and first-year running costs, with vendor-run IT, no general-trade outlet data, fixed plant capacity in summer, and a stakeholder approving every recommendation it makes. The model should cover all five areas at a high level, with depth on the two modules you choose under Objectives. No dataset is provided: build sample data from the figures in this case, which contains all the information you need. The focus is on how you identify the problems and reason through solving them, not on a polished system design.

Note: You may use vibe-coding tools or AI agent builders to create the model or agent.

Objectives

Choose exactly two of the modules below and show how they are linked: how a problem in one module caused or worsened problems in the other, and vice versa. For those two modules, identify the problems from the case facts, trace them to their root causes, and show how the model or agent would detect and mitigate the related risks across both. The solution must be resilient to disruptions, scalable to more flavors, depots and channels, integrated across the five areas, and efficient in cost and effort. Base each answer on the facts in this case; it contains all the information needed, so no outside assumptions are required. The case does not prescribe a solution. • Demand: What went wrong in demand planning last summer, and how should the model catch the next shift in demand early?
• Supply: What left supply exposed last summer, how large is the risk for next summer, and how should the model flag and respond to it?
• Inventory: Why did some flavors stock out while others were written off, and how should the model prevent both next summer?
• **Distribution:** Why did stock end up in the wrong places, and how should the model get it to where demand is?
• Transportation: What made inbound logistics fragile last summer, what did it cost, and how should the model reduce that exposure?

Deliverables Expectation

• Problem identification: The key problems in your two chosen modules, traced to their root causes, showing the cause-and-impact link in both directions: how problems in the first module affected the second, and how problems in the second affected the first, with last summer's loss quantified and attributed to those problems.
• Working model or agent: A working prototype (spreadsheet, script or low-code tool) that gives an end-toend view, with depth on your two chosen modules and the links between them, and flags risks with recommended actions. Show how it is resilient, scalable, integrated and efficient.
• Scenario test: A replay of last summer's viral spike and highway closure, as they affected your two chosen modules, showing what the model would have

flagged, when, and what it would have recommended. • Data and logic: The case data and decision rules behind the model, the trade-offs you made, and where a human makes the final call.
• Impact and next steps: The expected impact on next summer within the ₹2.5 crore budget, and what you would test or build next.

Submission

Teams of up to four have 42 hours to submit a deck of no more than 10 slides, including opening and closing slides, and the working prototype. Each team then presents for 10 minutes, followed by 5 minutes of questions from the judges.

Evaluation Criteria

Dimension	Weight	Description
Model or Agent Build	25%	How well the model gives end-to-end visibility, flags risks early and supports mitigation decisions.
Strategic Understanding	30%	Clarity in identifying the business problems and tracing them to their root causes.
Creativity	20%	Original, practical ideas for how the model detects risks and supports decisions.
Feasibility	15%	Use of the case data, and realism of timeline, capabilities, adoption and budget.
Presentation Clarity	10%	A clear, logical storyline from problem to solution.